

Skills Summary

Perimeter and Area of Coordinate Polygons

Use this reference while completing your worksheet or when studying.

Skill 1 — Perimeter from coordinates

Rule. Perimeter is the sum of the side lengths, and every side length comes from the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Two shortcuts: a *horizontal* side's length is the difference of the x -values, and a *vertical* side's length is the difference of the y -values — no square roots needed.

Example. Find the perimeter of the triangle with vertices $(1, 1)$, $(7, 1)$ and $(7, 9)$.

Step 1. Perimeter means every side, so take them one at a time — two of these sides are horizontal and vertical, so only the slanted one needs the full formula.

Step 2. horizontal side 6, vertical side 8, slanted side $\sqrt{6^2 + 8^2} = 10$; adding, $6 + 8 + 10$
perimeter = **24** units

Try it. Find the perimeter of the triangle with vertices $(2, 2)$, $(2, 7)$ and $(14, 7)$.

check: 24 units

Skill 2 — Area by the shoelace method

Rule. List the vertices once around the boundary in order, then returning to the first. Multiply each x by the *next* y and subtract each next x times the current y ; add all those terms and halve the size of the total:

$$\text{area} = \frac{1}{2} \left| \sum (x_i y_{i+1} - x_{i+1} y_i) \right|.$$

Going round the wrong way only changes the sign, which the size bars remove.

Example. Find the area of the quadrilateral $(0, 0)$, $(6, 0)$, $(8, 5)$, $(2, 5)$.

Step 1. The sides are not all horizontal or vertical, so there is no easy rectangle here — the shoelace sum handles any polygon given in order.

Step 2. terms 0, 30, 30, 0 add to 60, and half of 60 is the area
area = **30** square units

Try it. Find the area of the quadrilateral $(1, 0)$, $(9, 0)$, $(7, 4)$, $(3, 4)$.

check: 24 square units

Skill 3 — Area by splitting the shape

Rule. A polygon whose sides are all horizontal or vertical can be cut into rectangles: find each rectangle's area from its width and height, then add. This is the fastest check on a shoelace answer, because the two methods share no arithmetic — if they agree, both are almost certainly right.

Example. Find the area of the L-shaped hexagon $(0, 0)$, $(6, 0)$, $(6, 3)$, $(3, 3)$, $(3, 6)$, $(0, 6)$.

Step 1. Every side runs along a grid line, so cutting into rectangles is easier than the shoelace sum — one horizontal cut at height 3 does it.

Step 2. bottom rectangle $6 \times 3 = 18$, upper-left rectangle $3 \times 3 = 9$, and $18 + 9$ is the total area = **27** square units

Try it. Find the area of the L-shaped hexagon $(0, 0)$, $(10, 0)$, $(10, 2)$, $(4, 2)$, $(4, 6)$, $(0, 6)$.

check areas **98** square units

(!) Watch out: equal side lengths do not make a square. A rhombus with four sides of 5 has area 15, not 25 — only a rectangle's area is the product of two adjacent sides.